

# Paper Directions 16 — my read (Aug 4, 2026)

Source: [transcripts/20260804\\_Paper\\_Directions\\_16\\_otter\\_ai\\_transcript.docx](#) (~31 min).  
Context: Charles briefed Alex on Phase B characterization deck (intro +  $\rho$ ,  $\lambda$ ,  $\theta$ ,  $\theta \times K/N$ ,  $\gamma$ ,  $\lambda_{crit}$ ).  
Whiteboard: two sketches saved under [.cursor/.../assets/](#) (L\_C vs  $\rho$  distributions; naive-draft  $\theta$  from  $K/N$ ).

## Headline

**Strong meeting.** Alex liked the deck structure, sigmoid understanding, and sort-and-chop rationale for  $\gamma / \lambda_{crit}$ . The big pivot is **principled  $\theta$  from  $K/N$**  (naive draft) and **reframing  $\rho + L_C$  via team-level congestion distributions** — not just selection-curve OAT plots.  
Closing energy: move from “539 preset knobs” toward **data-identified**  $\rho$ ,  $\theta$ , (then principled  $\gamma$ ,  $\lambda$ ) — only  $\gamma$  and  $\lambda$  remain free after  $\theta$  is fixed from ability draw +  $K/N$ .

## What landed well

- Intro / glossary / benchmarks —  $A_i$ ,  $T_j$ , smooth  $L_C$ , seed 42, not curve-fitting.
- $\rho$  slides — “Assortativity matters” accepted; Alex wants **deeper intuition** (see gaps).
- $\lambda$  slide — talent-only vs  $\lambda > 0$  bend; same story as before.
- $\theta \times K/N$  — co-movement readout; Alex said continue, revisit  $\theta$  interpretation.
- $\gamma / \text{sort-and-chop} / \lambda_{crit} \approx 4/\gamma$  — Charles’s “no ambiguity, deterministic” explanation accepted; Alex not demanding full derivation now.

## Whiteboard ↔ transcript (capture for slides / memo)

### Sketch A — distribution of $L_C$ over teams vs $\rho$

| Panel           | Axes                                                                    | Low $\rho$                                                                 | High $\rho$                                                                                                         |
|-----------------|-------------------------------------------------------------------------|----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| 1D (top/middle) | $x = L_C$ , $y = \# \text{ teams}$                                      | Single <b>narrow hump</b> (Alex: nearly a <b>delta</b> , not even uniform) | <b>Spread</b> / bimodal — weak teams $L_C \approx 0$ , strong teams $L_C$ high                                      |
| 2D (bottom)     | $x = L_C$ , $y = \text{team ability (mean } A \text{ on team or } T_j)$ | —                                                                          | Heatmap: <b># teams</b> ; expect <b>upward cloud</b> — <b>better teams <math>\rightarrow</math> more congestion</b> |

**Alex lock:**  $\rho$  dials the spread of the  $L_C$  distribution across teams — practical interpretation of assortativity in *this* model, not only “selection curve bends.”  
Charles linked this to sort-and-chop picture (elite vs weak team  $L_C$ ) — Alex: “not coincidence.”

### Sketch B — naive draft $\rightarrow$ principled $\theta$

- Draw  $A_i$  from ability distribution (Beta on  $[0,1]$  in sim).
- No teams** — everyone competes on ability alone; top  $K/N$  advance.
- Find cutoff  $A^*$  such that  $P(A > A^*) = K/N$  (area under PDF to the right of cutoff).
- That cutoff defines  $\theta$**  (viability / ability threshold).

**Meaning (Alex):**  $\theta$  is the ability level where, **if there were no roster ambiguity**, everyone above would advance and everyone below would not. The sigmoid then adds **ambiguity around that threshold** (peer viability graded, not cliff).  
**Not the same as today’s code default:** fixed  $\theta = 0.72$  from [tier1\\_539\\_reference\\_settings.json](#). PD16 direction = **compute  $\theta$  from  $K/N$  + ability draw** each league/domain.

## Model changes Alex asked to try (priority order)

- $L_C$  per team (not LOO per player)**
  - Congestion = property of **team  $j$** , same for all players on roster.
  - Alex: earlier hero-style runs used **poolq** (not LOO) and worked; LOO may be unnecessary for  $L_C$  in score.
  - Action:** implement team-level  $L_C$ ; verify Pass B curve + Army path unchanged “in the wash.”
- $\theta$  from  $K/N$  + ability CDF**
  - Replace free  $\theta$  preset with **quantile**:  $\theta = F_{A^{-1}}(1 - K/N)$  on the draw used for  $A_i$  (Beta or empirical).
  - Action:** helper + re-run  $\theta$ /OAT and  $\theta \times K/N$  with computed  $\theta$ ; document shift from 0.72.
- $\rho \leftrightarrow L_C$  distribution diagnostic (new figure / slide gap)**
  - For each  $\rho$  arm: histogram of **team  $L_C$**  (# teams vs  $L_C$ ).
  - 2D heatmap: team mean ability (or  $T_j$ ) vs  $L_C$ .
  - Action:** new script or Pass C extension; fills whiteboard Sketch A.
- $\rho$  low arm still not flat enough**
  - Try  **$\rho \rightarrow 0.001$** ; check bin count / stochastic assignment.
  - Build **one clean sentence** + empirical/analytic intuition for why  $\rho$  matters in *this* construction.
- $\gamma$  and  $\lambda$  — principled fit (later)**
  - After  $\rho$ ,  $\theta$  data-linked: only  $\gamma$ ,  $\lambda$  free.

- Alex: avoid “make sim curve match hero” as the *definition*; find **data properties** that pin  $\gamma$ ,  $\lambda$ .

**Parked:** Menger **multiplicative** congestion (Charles → 4-sentence memo); additive mean vs product.

### Empirical calibration roadmap (Alex, end of meeting)

| Knob                        | From data? | How                                          |
|-----------------------------|------------|----------------------------------------------|
| A <sub>i</sub> distribution | Yes        | Empirical perf or declared draw (Beta)       |
| K/N                         | Yes        | Domain (MBB ~1%, etc.)                       |
| $\theta$                    | Yes (PD16) | Quantile from A distribution at $1 - K/N$    |
| $\rho$                      | Yes        | Match assignment assortativity in data       |
| $\gamma$ , $\lambda$        | Open       | Need principled estimators — not preset-only |

### Sort-and-chop (what Charles told Alex)

Used for  $\gamma / \lambda_{\text{crit}}$  slides only: removes assignment ambiguity → clean  $\lambda_{\text{crit}} \approx 4/\gamma$  story. Alex OK’d; wanted congestion unpacked separately (team L<sub>C</sub>,  $\theta$  from K/N) — not a rejection of sort-and-chop.

### Quotes to keep

- **$\rho$ :** “Assortativity is this dialing measure of what’s the spread of the L<sub>C</sub>s.”
- **$\theta$ :** “In a world where there was no team congestion, the cutoff for ability would have been ... pick  $\theta$  so that area = K/N.”
- **L<sub>C</sub>:** “Congestion should be how many good players are on the team.”
- **Fit:** “Try to find  $\gamma$  and  $\lambda$  without saying the way to do it is just make the two curves as close as possible.”

### Suggested next COMPASS / Charles actions

**Live checklist:** CHARLES\_CHECKLIST.md (post-PD16). Archived: archive/checklists/.

- ✓ Draft PD16 one-pager for re\_entry/ → 08\_PD16\_Alex\_meeting\_takeaways.md
- ✓ Refresh Charles checklist for PD16 priorities
- ☐ Implement team L<sub>C</sub> +  $\theta(K/N)$  via shell env vars ( GALLERY\_LC\_MODE , GALLERY\_THETA\_MODE in gallery\_knobs.py ) — **not** a new conda env; compare PNGs to current deck
- ☐ Build L<sub>C</sub> distribution vs  $\rho$  figures (whiteboard Sketch A)
- ☐  $\rho \rightarrow 0.001$  arm + assortativity intuition paragraph for slide footer
- ☐ Menger multiplicative — 4 sentences (parked)
- ☐ Update intro slide when  $\theta$  provenance changes (539 preset → K/N quantile)

**Related:** Phase B walkthrough 07\_Phase\_B\_Characterization\_Slides\_Explained.md ; memo 06 ( $\lambda_{\text{crit}}$ ); PD15 (characterization vs fitting).